HAPPY DIWALI · PYTHON · PANDAS

Diwali Sales Analysis



Who buys, where they live, and what they buy during India's festival of lights.

Business Analytics Project | Data Analytics Portfolio

₹10.62 Cr	27,981	3,752	₹9,454
Total sales	Orders placed	Unique customers	Avg. per record

Project overview

This project analyses Diwali festive-season sales to understand customer behaviour and find which segments, regions and products drive the most revenue.



Customer profile

Compare spending by gender, age group and marital status.



Geography

Find the states that generate the most orders and sales.



Occupation

See which professions have the highest purchasing power.



Products

Identify top categories and the most-ordered products.

Dataset overview and cleaning

Dataset: 13 columns

Group	Columns
Customer	User_ID, Cust_name, Gender, Age, Age Group, Marital_Status
Location	State, Zone
Work	Occupation
Purchase	Product_ID, Product_Category, Orders, Amount

11,239 clean rows used for analysis

Data cleaning steps

Dropped 2 empty columns

Status and unnamed1 had no data.

Removed 12 null rows

Missing Amount values: 11,251 to 11,239 rows.

Fixed data type

Amount converted from float to int.

Checked duplicates

8 repeated rows flagged during review.

Q1. Who spends more: men or women?

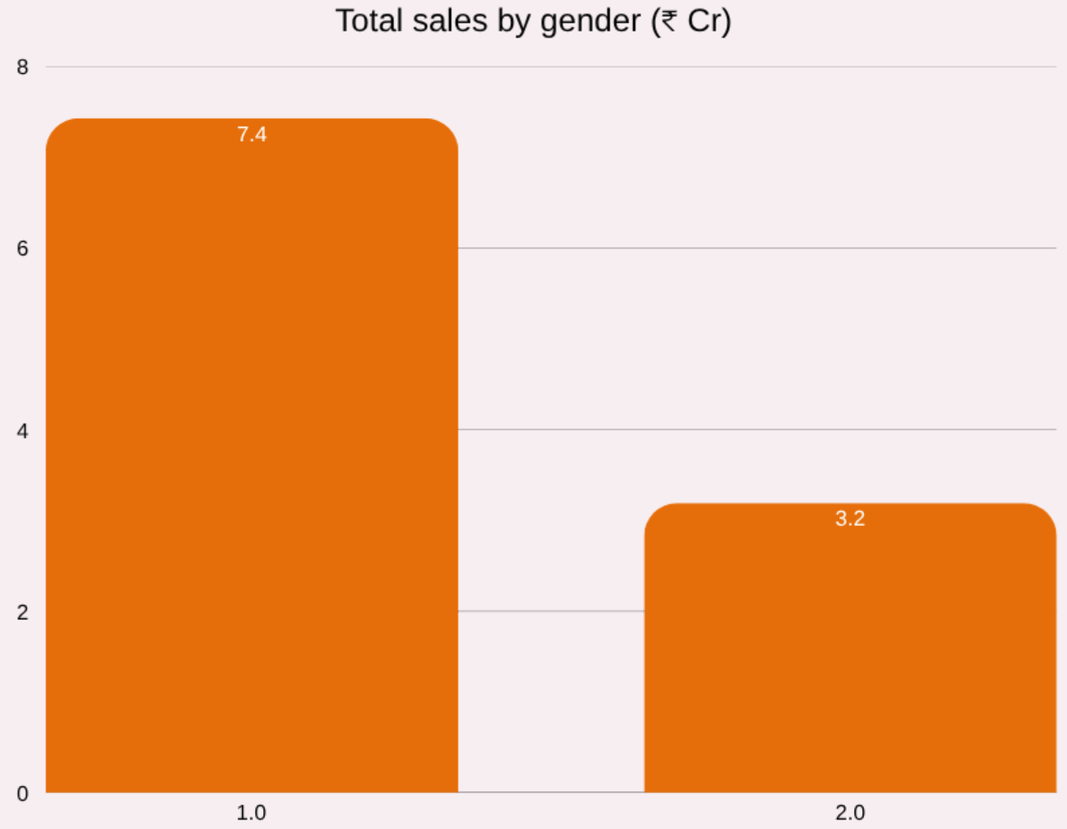
PANDAS

```
gender_sales = (df.groupby("Gender",
    as_index=False)["Amount"].sum()
    .sort_values("Amount",
                 ascending=False))

sns.barplot(x="Gender", y="Amount",
            data=gender_sales)
```

70%

of total sales come from women, who also place most orders



Q2. Which age group drives the most sales?

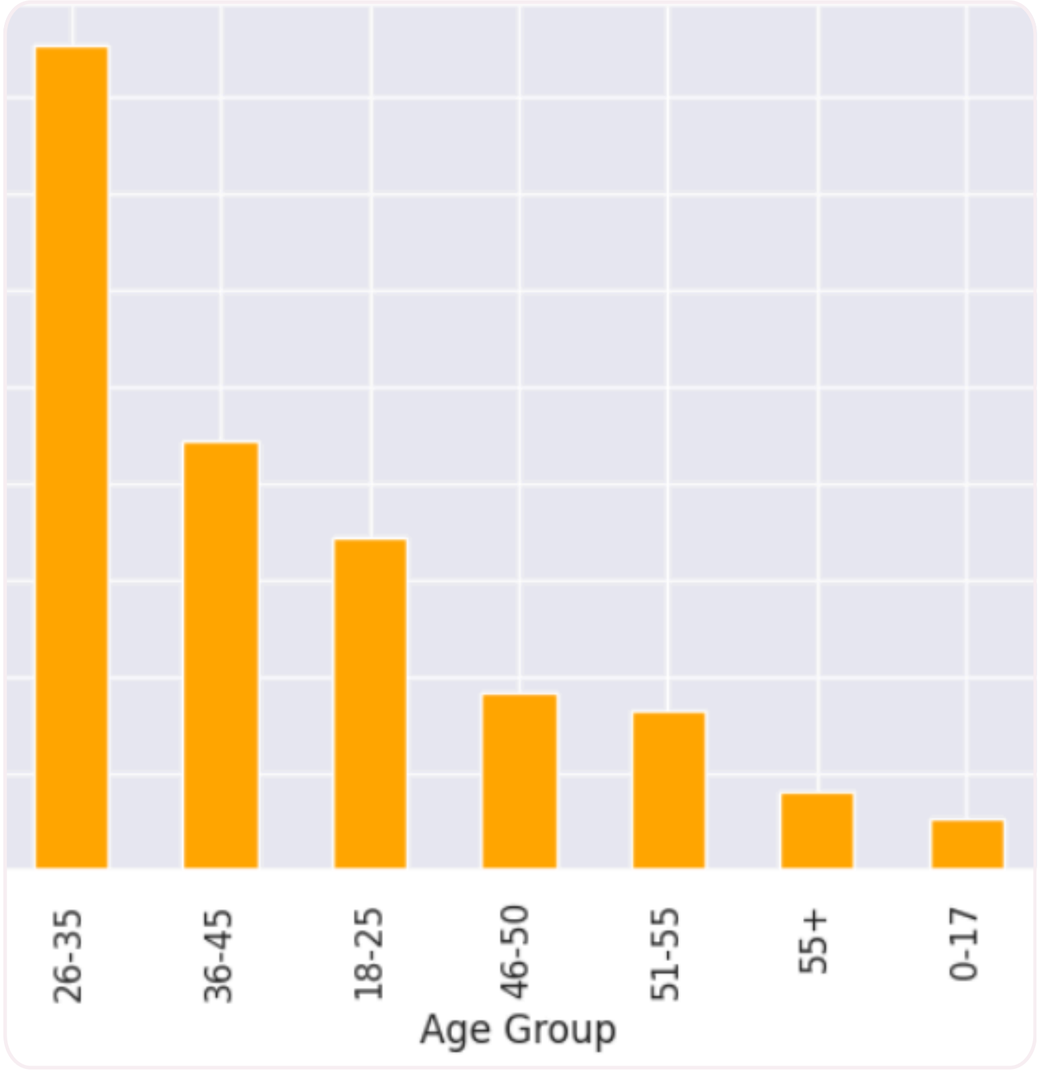
PANDAS

```
age_sale = (df.groupby("Age Group",
    as_index=False)["Amount"].sum()
    .sort_values("Amount",
                 ascending=False))

sns.barplot(x="Age Group", y="Amount",
            data=age_sale)
```

40%

of sales come from the 26-35 age group, the biggest segment



Q3. Which states contribute the most sales?

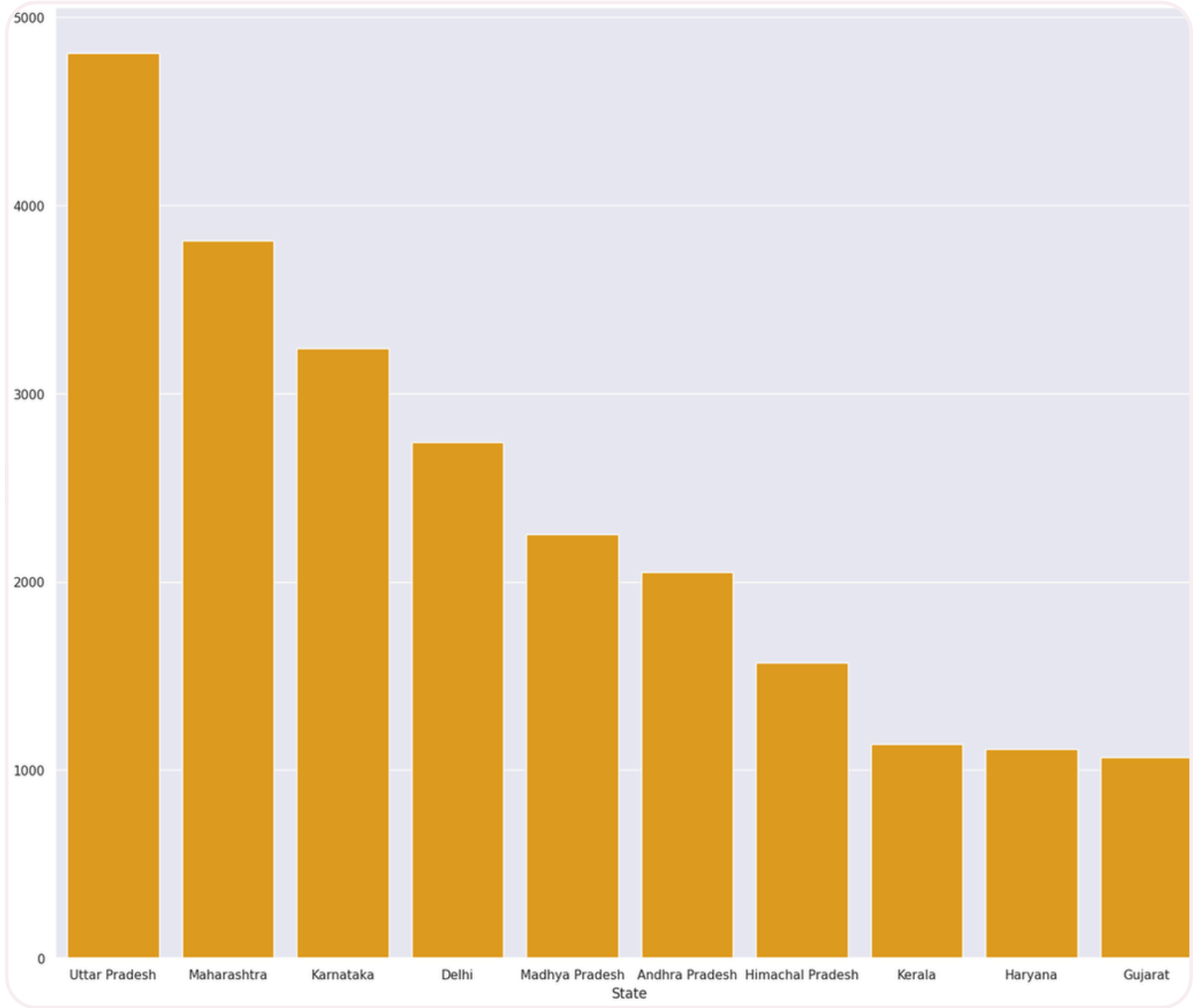
PANDAS

```
state_by_amount = (df.groupby("State",
    as_index=False)["Amount"].sum()
    .sort_values("Amount",
                 ascending=False)
    .head(10))

sns.barplot(y="Amount", x="State",
            data=state_by_amount)
```

55%

of sales come from just 4 states: UP, Maharashtra, Karnataka, Delhi



Q4. How do marital status and gender affect spending?

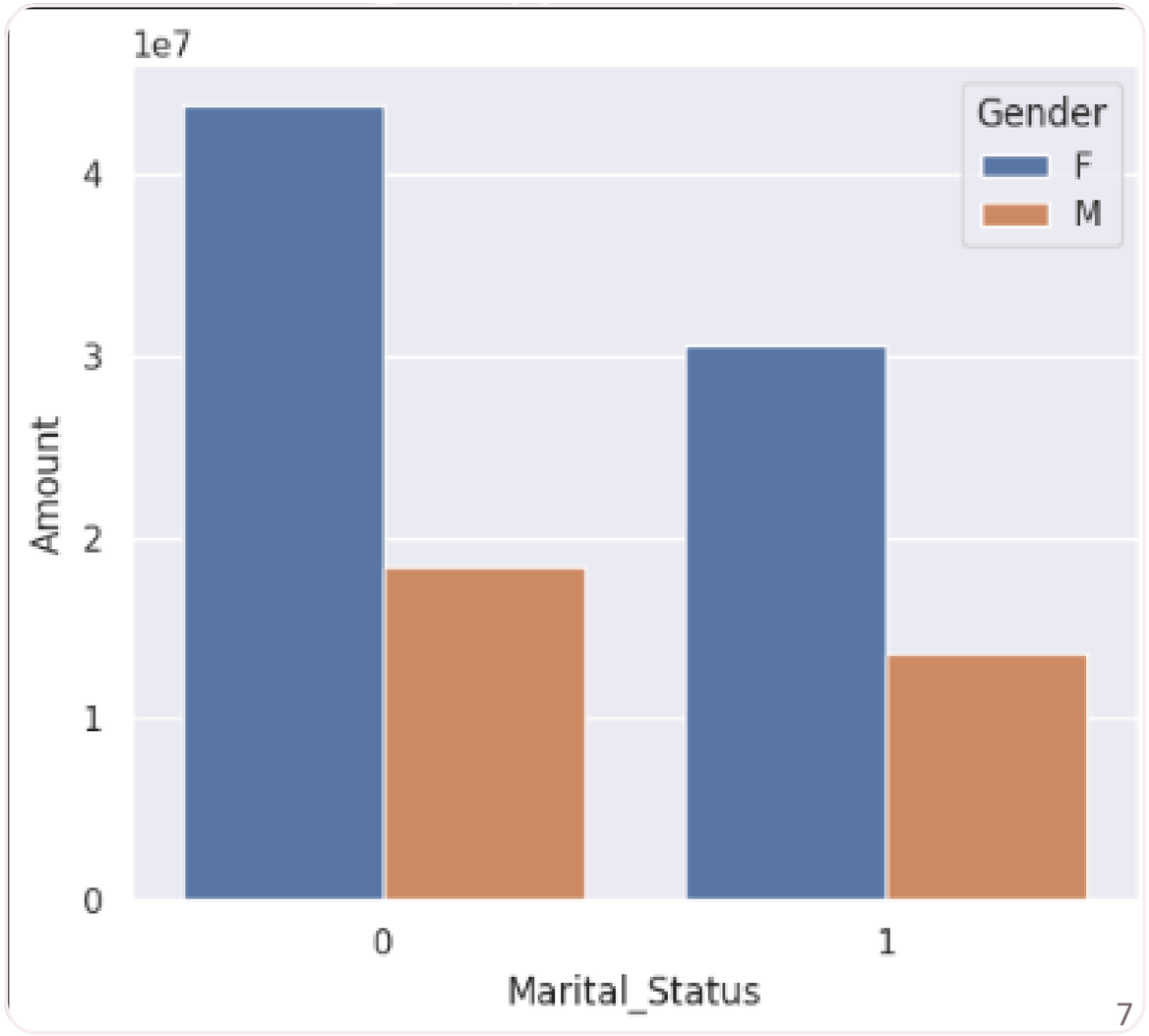
PANDAS

```
marital_by_amount = (df.groupby(
    ["Marital_Status", "Gender"],
    as_index=False)["Amount"].sum())

sns.barplot(x="Marital_Status",
            y="Amount", hue="Gender",
            data=marital_by_amount)
```

₹4.38 Cr

spent by unmarried women, the top-spending segment

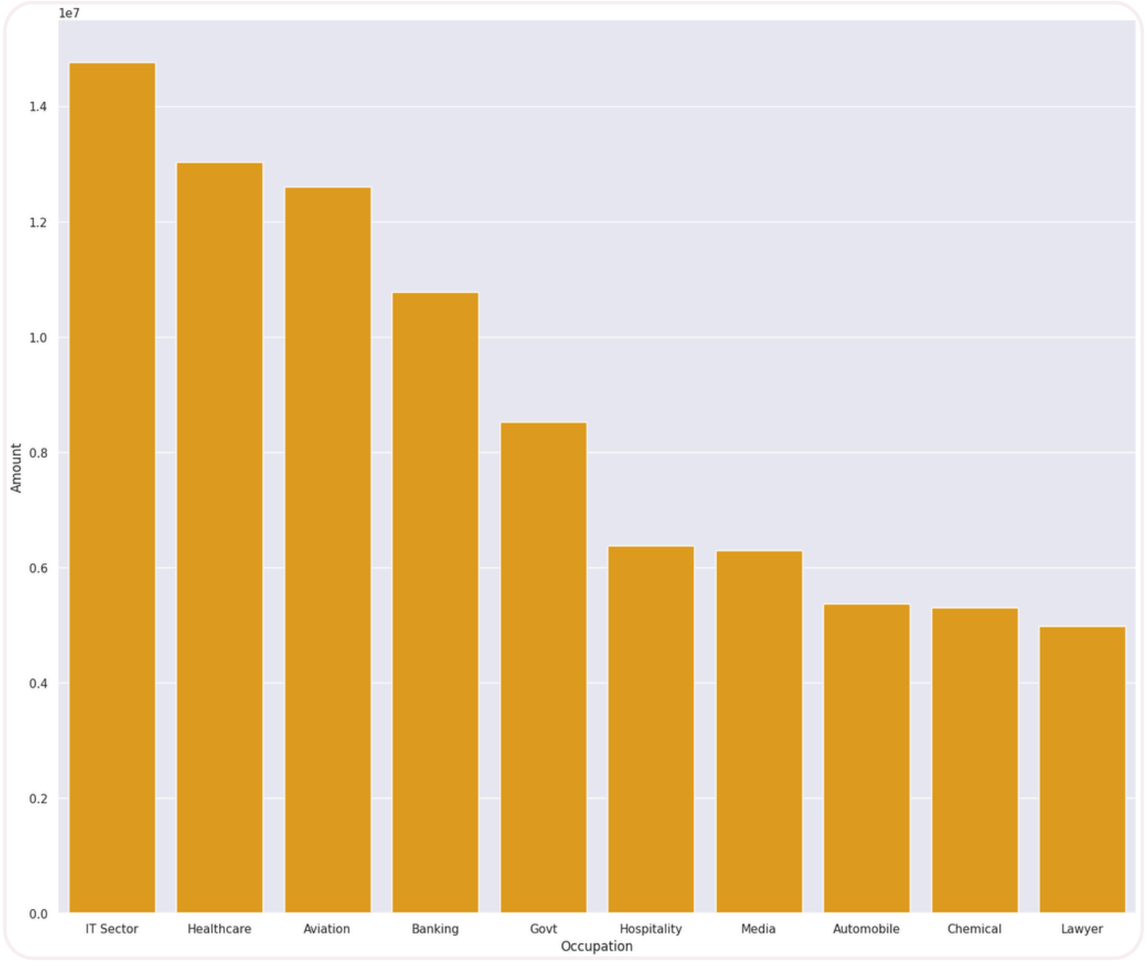


Q5. Which occupations spend the most?

PANDAS

```
occ_by_amt = (df.groupby("Occupation",
    as_index=False)["Amount"].sum()
    .sort_values("Amount",
                 ascending=False)
    .head(10))

sns.barplot(x="Occupation", y="Amount",
            data=occ_by_amt)
```



38%

of sales come from IT Sector, Healthcare and Aviation

Q6. Which product categories sell the most?

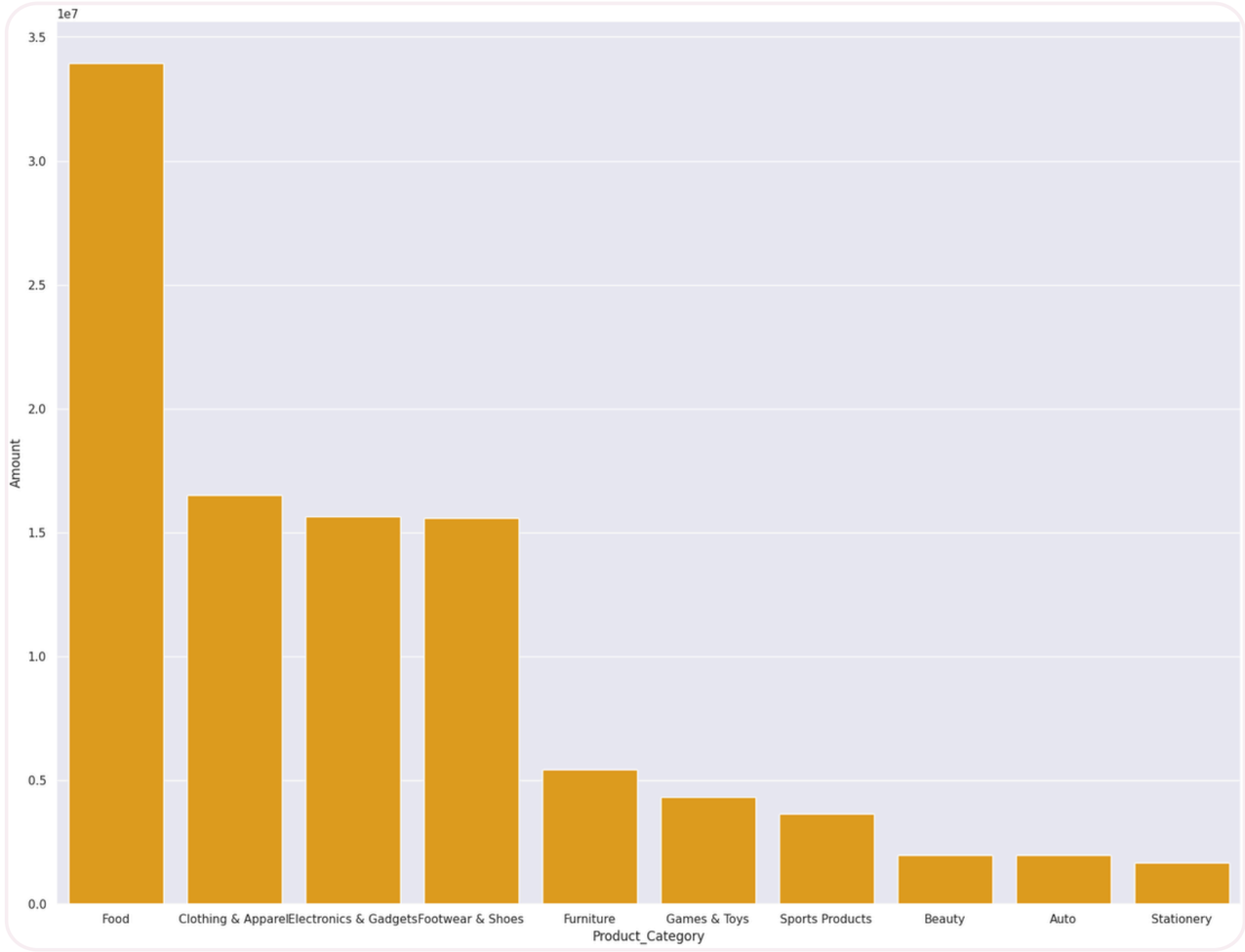
PANDAS

```
pro_amt = (df.groupby("Product_Category",
    as_index=False)["Amount"].sum()
    .sort_values("Amount",
                 ascending=False)
    .head(10))

sns.barplot(x="Product_Category",
            y="Amount", data=pro_amt)
```

32%

of sales are Food, nearly double the next category



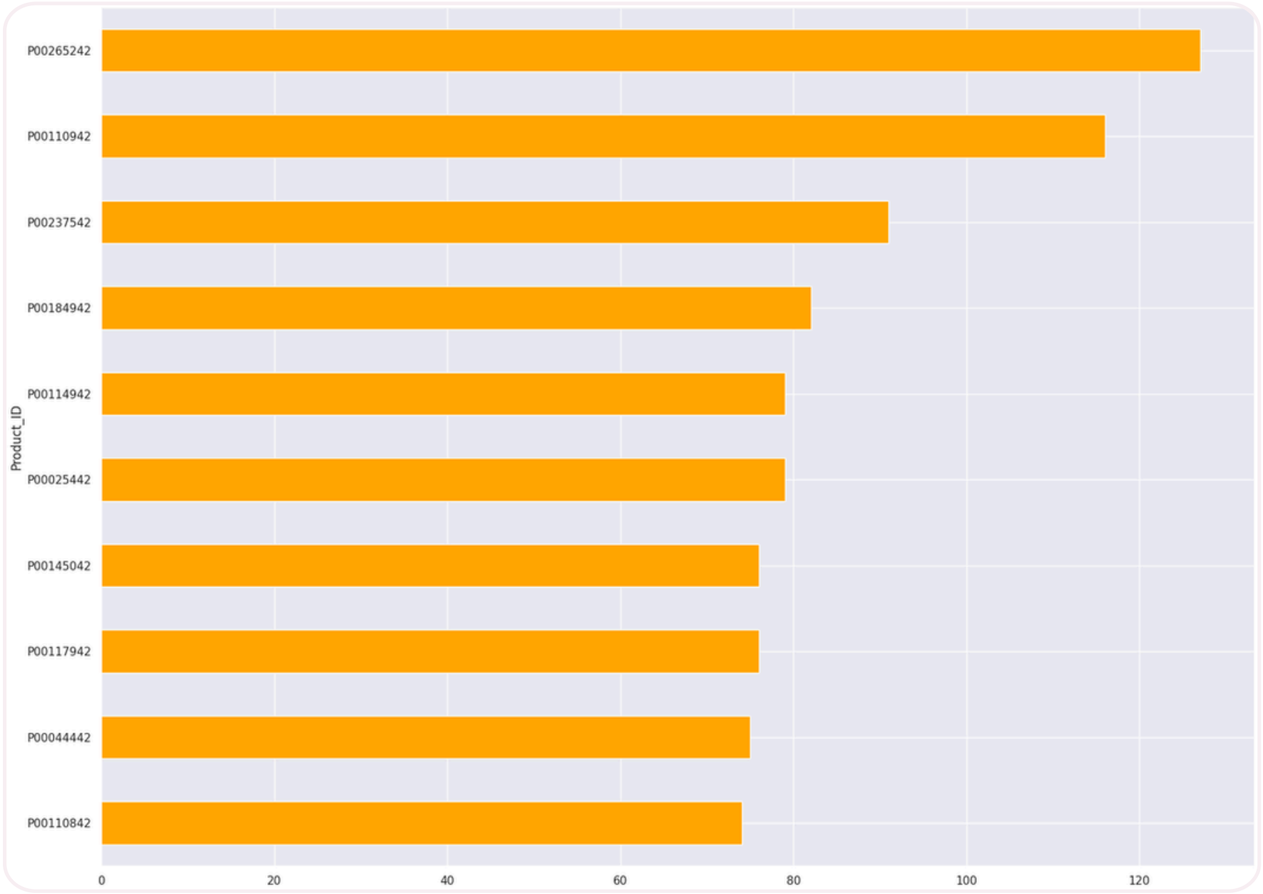
Q7. Which products get the most orders?

PANDAS

```
(df.groupby("Product_ID")["Orders"]
   .sum()
   .nlargest(10)
   .sort_values(ascending=False)
   .plot(kind="barh"))
```

127

orders for the best seller, product P00265242



Key insights



Women lead

Women account for 70% of sales, and ages 26-35 are the biggest buyers.



Four states dominate

Uttar Pradesh, Maharashtra, Karnataka and Delhi bring in over half of sales.



Professionals spend

IT Sector, Healthcare and Aviation workers have the highest spend.



Festive essentials win

Food, Clothing, Electronics and Footwear make up 77% of sales.

Conclusion and recommendations

Top buyer profile

Women aged 26-35 from Uttar Pradesh, Maharashtra and Karnataka, working in IT, Healthcare or Aviation, who mostly buy Food, Clothing and Electronics.

1

Target women 26-35

Focus festive campaigns and offers on this core segment.

2

Invest in top states

Prioritise stock and ads in UP, Maharashtra, Karnataka and Delhi.

3

Stock festive essentials

Keep Food, Clothing, Electronics and Footwear well supplied.

Thank You!

We turned 11,239 festive transactions into a clear picture of who buys, where they live, and what they buy, using 7 targeted analyses in Python and Pandas.

This is how retail analysts plan festive inventory, marketing and regional targeting.

KEY RESULTS

₹10.62 Cr	27,981	3,752	70%
Total sales	Orders	Customers	Sales by women

BUSINESS IMPACT

- Targeted campaigns
- Regional focus
- Better festive stocking
- Higher festive sales

PRESENTED BY:

Kamal Damani

Data Analytics Portfolio

TOOLS & TECHNOLOGIES

- Python
- Pandas
- Seaborn

KEY TAKEAWAYS

- Actionable insights
- Strategic decisions
- Business growth

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